LAB REPORT – 10

Student Name	**MADDUKURI UMA**	
Student Registration Number	252U1R1141	**Class &Section:** CSE-2C & 213
Study Level : UG/PG	**UG**	***Year &Term*:** II & i
Subject Name	**C++**	

Name of the Assessment	**Lab Report - 10**
Date of Submission	**03-10-26**

LAB REPORT – MODULE 10

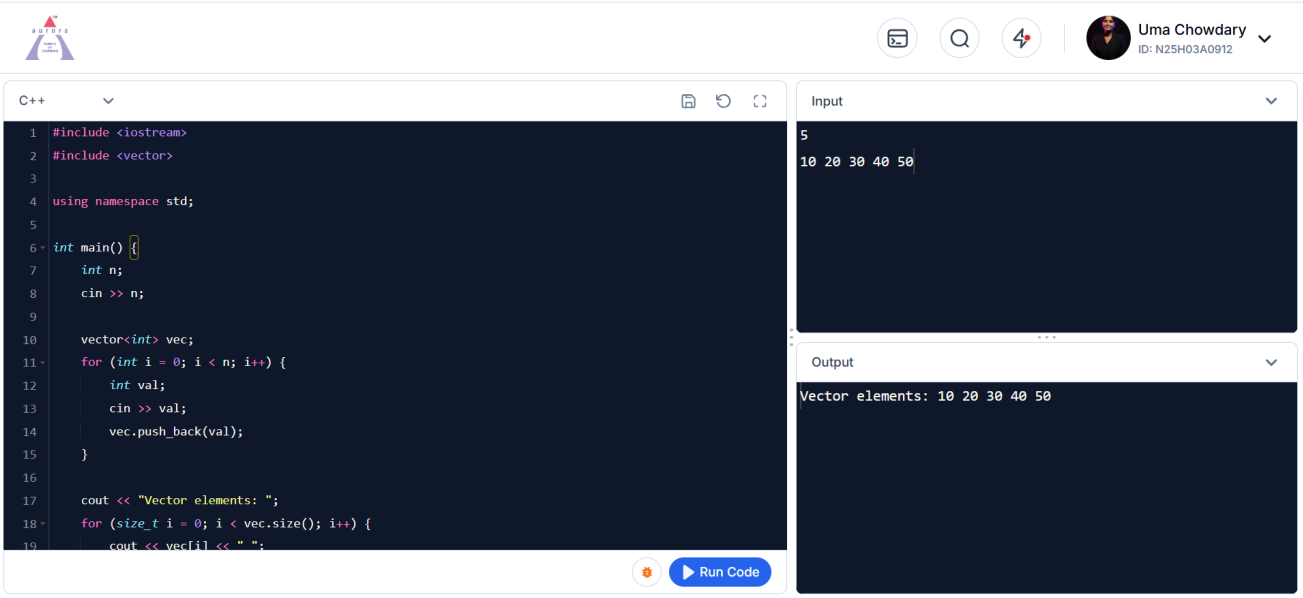
Standard Template Library (STL) in C++

Module 10 – Coding Questions

Question 1 – Vector: Insert and Display

1. Create a vector of integers. Read N integers from the user and insert them into the vector using push_back(). Display all the elements of the vector.

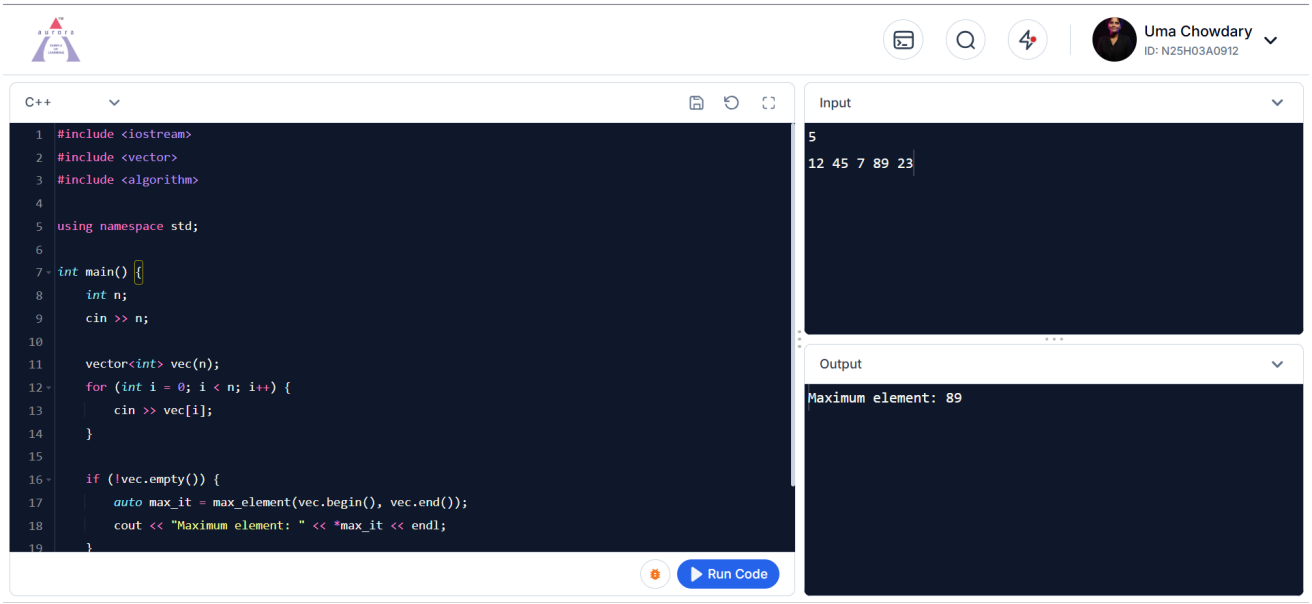
Concepts: vector, push_back(), size(), and iteration.



Question 2 – Vector: Find the Maximum Element

Create a vector of N integers. Read the elements and find the maximum element using the max_element() function. Display the maximum element.

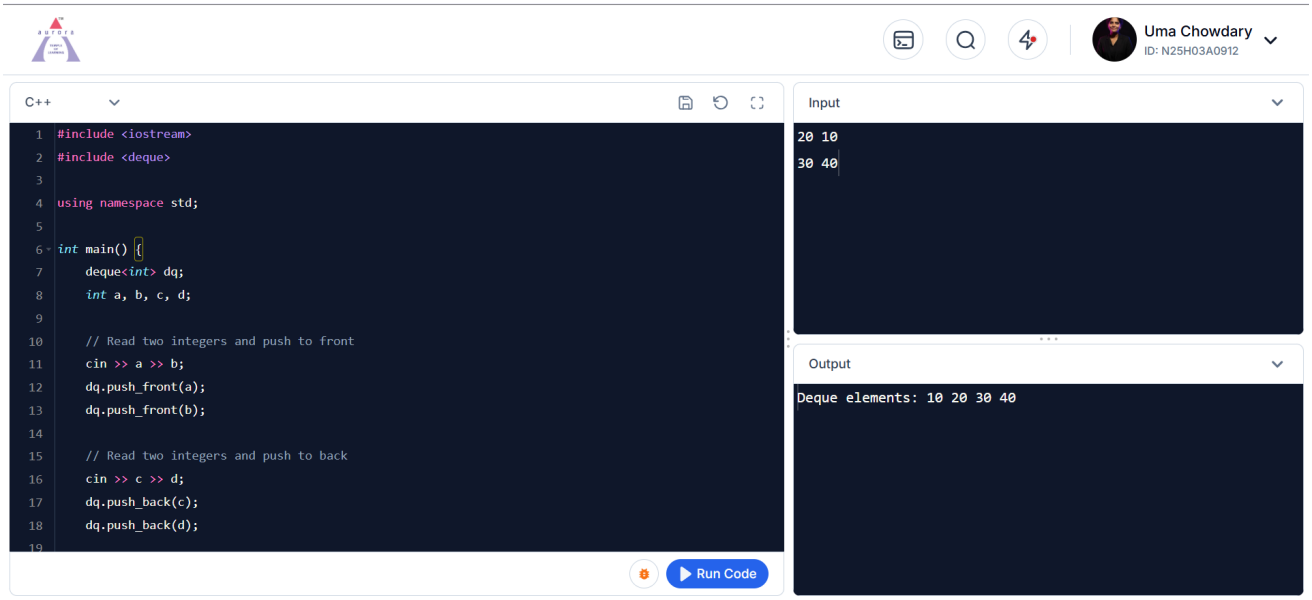
Concepts: vector and max_element().



Question 3 – Deque: Insert Elements at Both Ends

Create a deque of integers. Read two integers and insert them at the front using push_front(). Read two more integers and insert them at the back using push_back(). Display all the elements.

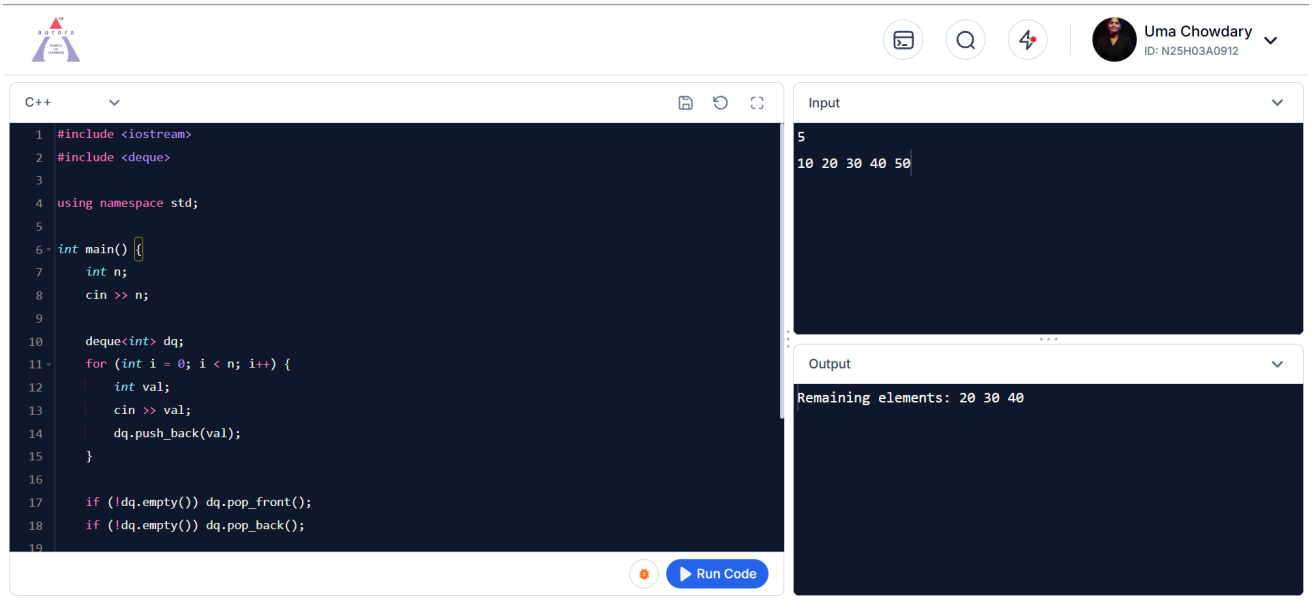
Concepts: deque, push_front(), push_back(), and traversal.



Question 4 – Deque: Delete Elements from Both Ends

Create a deque of integers and insert N elements into it. Remove the first element using pop_front() and the last element using pop_back(). Display the remaining elements.

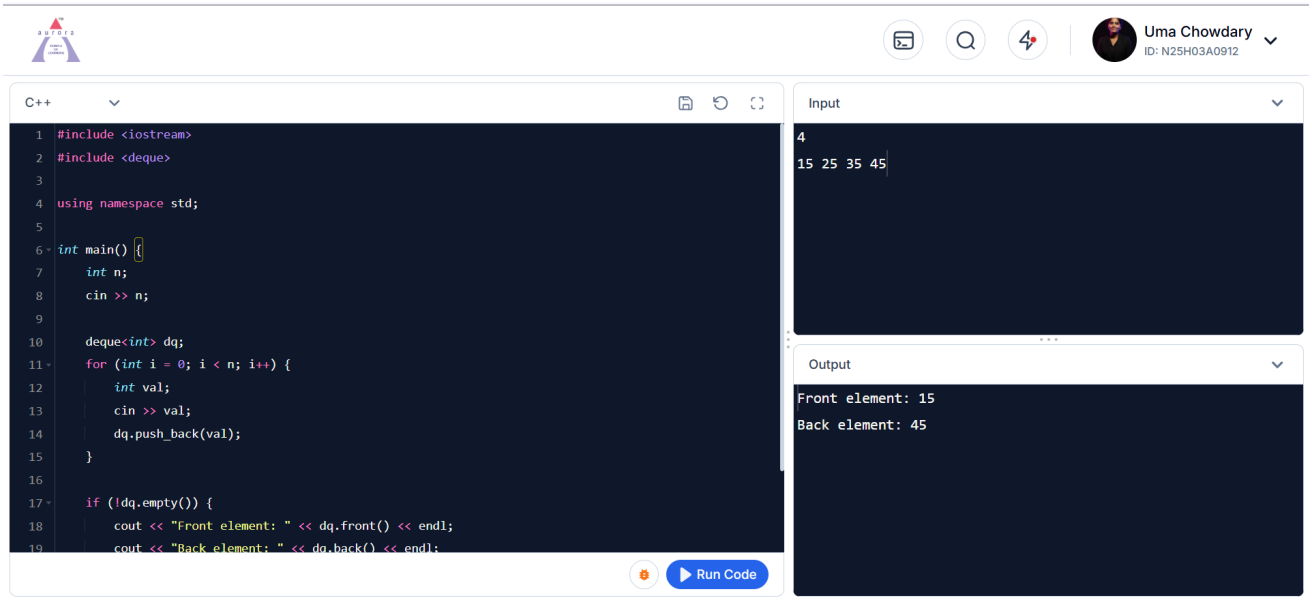
Concepts: deque, pop_front(), pop_back(), and empty().



Question 5 – Deque: Access Front and Back Elements

Create a deque of integers. Read N integers and insert them into the deque. Display the first and last elements using front() and back(), respectively.

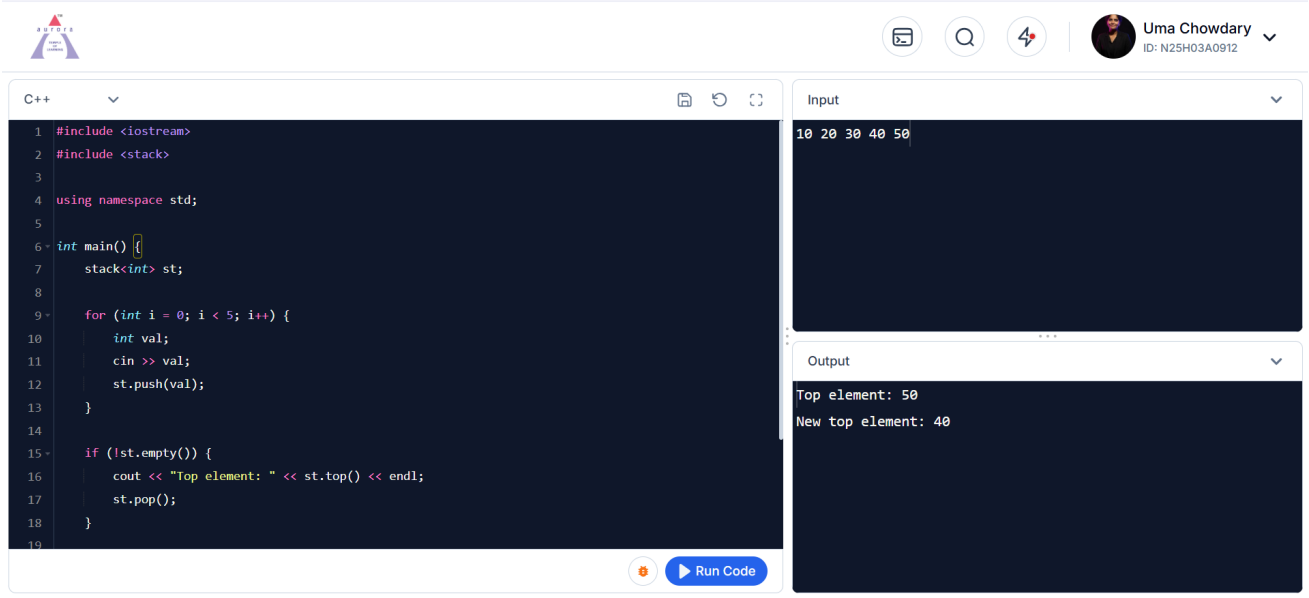
Concepts: deque, front(), back(), and size().



Question 6 – Stack: Basic Operations

Create a stack of integers. Push five integers into the stack. Display the top element, remove the top element using pop(), and display the new top element.

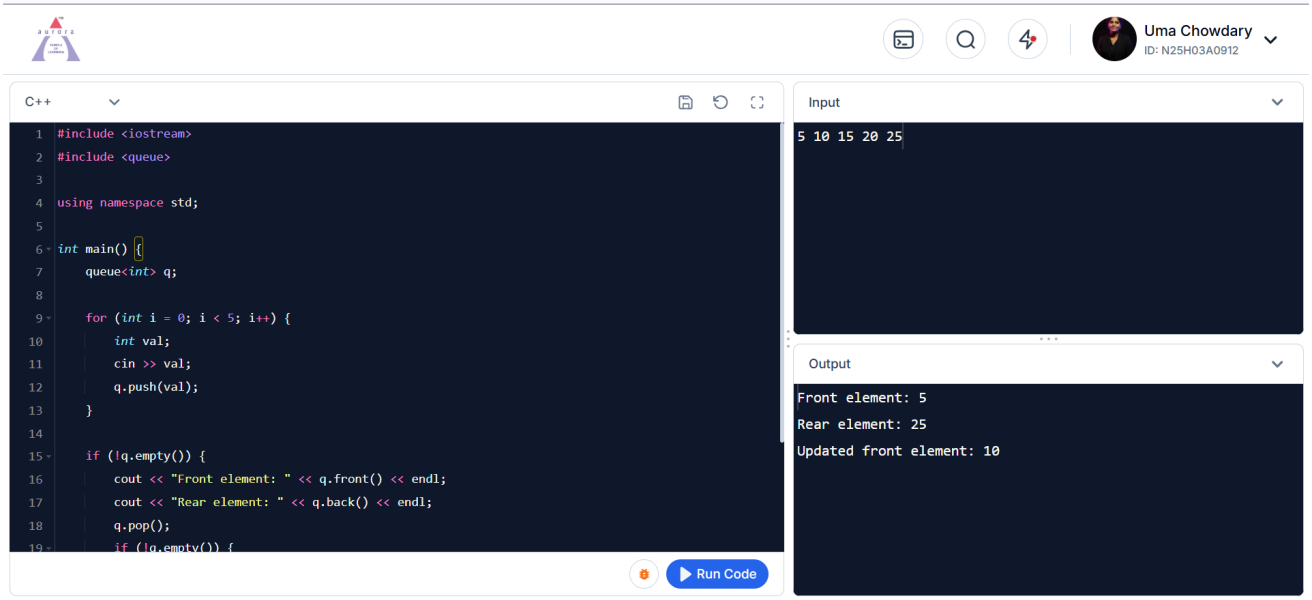
Concepts: stack, push(), pop(), top(), and empty().



Question 7 – Queue: Basic Operations

Create a queue of integers. Insert five integers using push(). Display the front and rear elements. Remove the front element using pop() and display the updated front element.

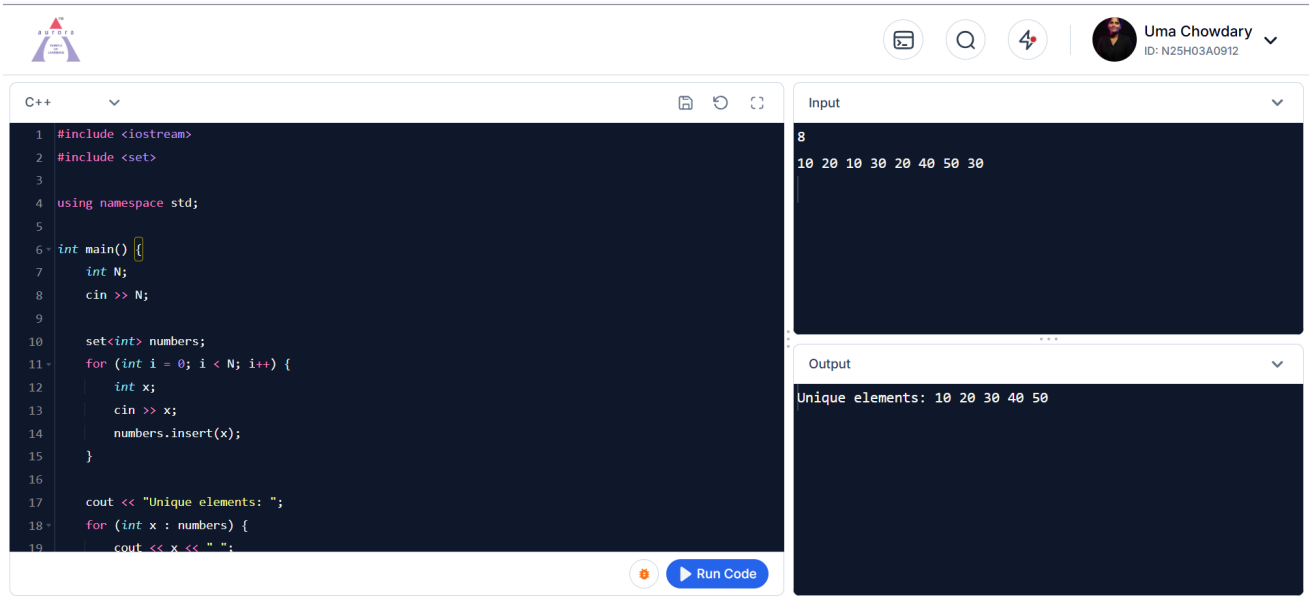
Concepts: queue, push(), pop(), front(), and back().



Question 8 – Set: Store Unique Elements

Create a set of integers. Read N integers from the user and insert them into the set. Display all the elements of the set.

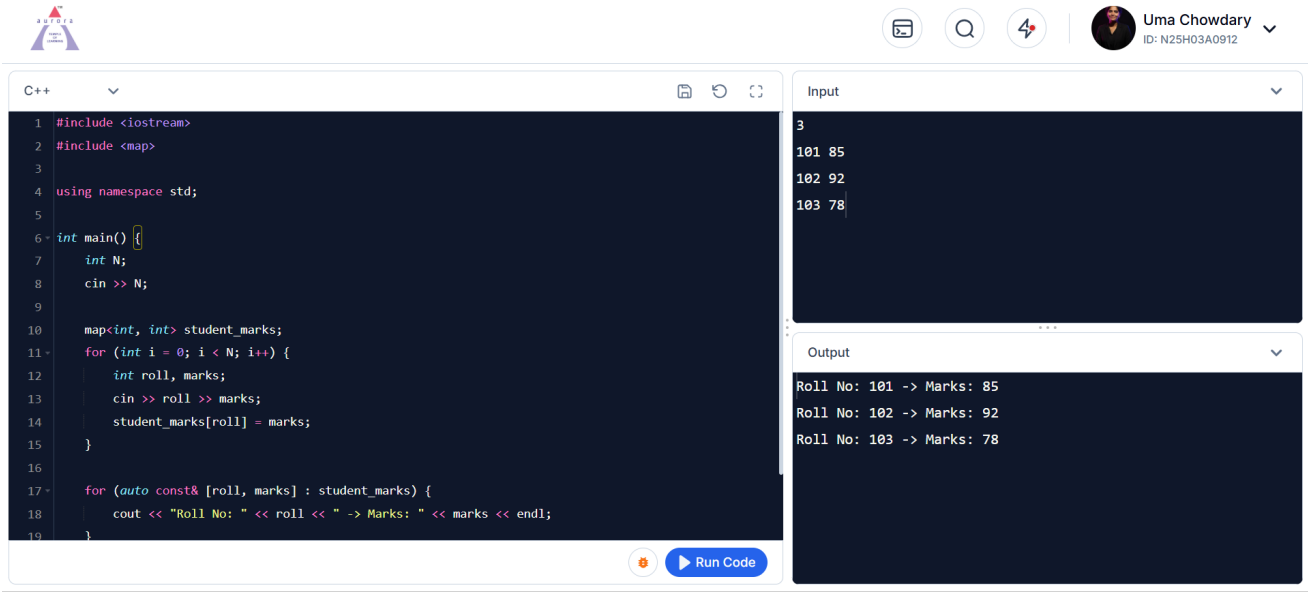
Concepts: set, insert(), iteration, and unique elements.



Question 9 – Map: Student Marks

Create a map in which the key is a student's roll number and the value is their marks. Read N pairs of roll numbers and marks, insert them into the map, and display all the key-value pairs.

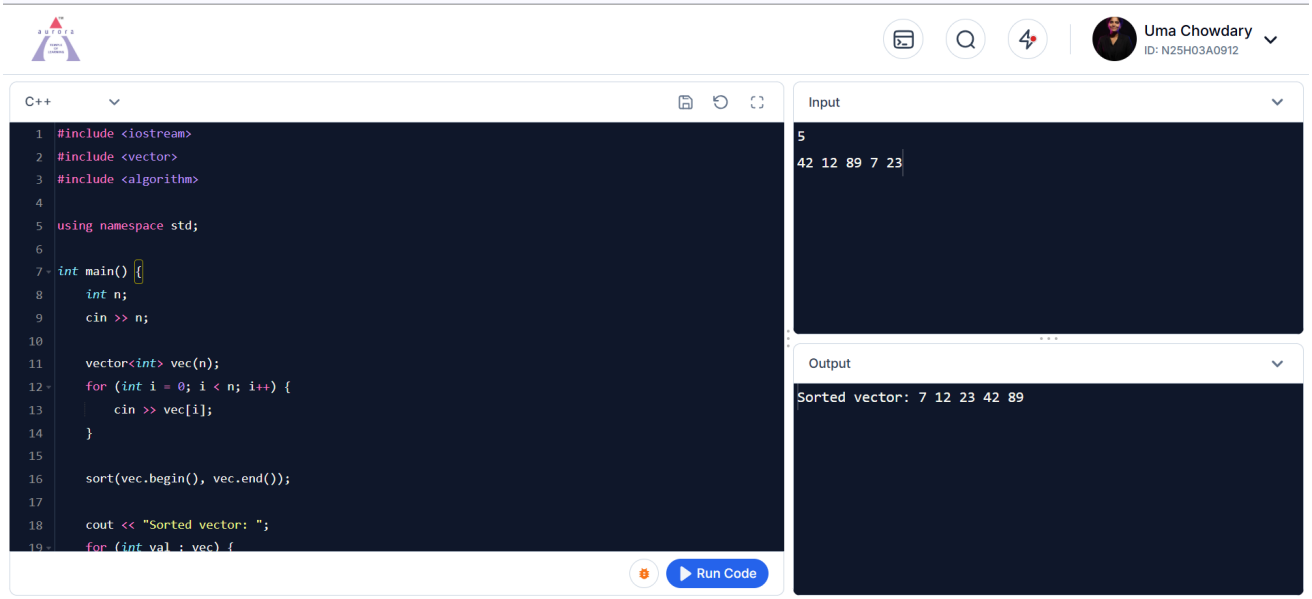
Concepts: map, insert(), key-value pairs, and iteration.



Question 10 – Sorting Using STL

Read N integers into a vector. Sort the elements in ascending order using the STL sort() function and display the sorted vector.

Concepts: sort(), vector, begin(), and end.



Conclusion

The programs in this module helped us understand and implement the Standard Template Library (STL) in C++. We learned how to use different STL containers and functions such as queue, set, map, vector, push(), pop(), front(), back(), insert(), and sort(). The programs demonstrated how STL makes data storage, processing, searching, and sorting easier and more efficient. Thus, the objectives of the module were successfully achieved by implementing and testing the given programs.

Submission Checklist

1. All 10 questions are completed.
2. Every program is written in the Coding Playground only.
3. The code has been successfully executed and tested.
4. A clear screenshot of each solution is attached.
5. Your profile picture and name are visible in every code screenshot.
6. The complete code is visible and readable.
7. No copied or shared solution has been submitted.

Note: A screenshot without the student's profile picture and name will be considered copied and will receive 0 marks.